

Curtis Fox

BASIC INFORMATION Email: curtfox@student.ubc.ca
[Google Scholar](#) | [Website](#) | [GitHub](#) | [LinkedIn](#)

SUMMARY PhD student in machine learning with a research focus on optimization and deep learning. Work accepted at ICML, ICLR, TMLR, IEEE Transactions on Information Theory, and NeurIPS OPT Workshop. Industry research experience developing evaluation benchmarks for large language models.

EDUCATION **University of British Columbia**
Doctor of Philosophy (PhD) in Computer Science *2023 - Present*

- Advisor: Mark Schmidt

Master of Science (MSc) in Computer Science *2021 - 2023*

- Advisor: Mark Schmidt
- Master's Thesis: *A Study of the Edge of Stability in Deep Learning*

Bachelor of Science (BSc) *2014 - 2019*

- Major: Combined Honours in Computer Science and Statistics

PAPERS

1. **Flatland: The Adventures of Gradient Descent with Large Step Sizes**
L. Galli*, **C. Fox***, W. Bartolomeaus, M. Schmidt, H. Rauhut. *ICML*, 2026 [[Link](#)]
2. **Glocal Smoothness: Line search and adaptive step sizes can help in theory too!**
C. Fox, A. Mishkin, S. Vaswani, M. Schmidt. *TMLR*, 2026 [[Link](#)][[NeurIPS OPT Workshop version](#)]
3. **DRBench: A Realistic Benchmark for Enterprise Deep Research**
A. Abaskohi, T. Chen, M. Muñoz-Mármol, **C. Fox**, A. V. Ramesh, É. Marcotte, X. H. Lù, N. Chapados, S. Gella, C. Pal, A. Drouin, I. H. Laradji. *ICLR*, 2026 [[Link](#)]
4. **Next-token Prediction Capacity: General Upper Bounds and a Lower Bound for Transformers**
L. Madden, **C. Fox**, C. Thrampoulidis. *IEEE Transactions on Information Theory*, 2025 [[Link](#)]
5. **Nonmonotone Line Searches Operate at the Edge of Stability**
C. Fox*, L. Galli*, M. Schmidt, H. Rauhut. *NeurIPS OPT Workshop*, 2024 [[Link](#)]
6. **A Study of the Edge of Stability in Deep Learning**
C. Fox. *Master's Thesis*, 2023 [[Link](#)]
7. **Deep Learning of Immune Cell Differentiation**
A. Maslova, R. N. Ramirez, K. Ma, H. Schmutz, C. Wang, **C. Fox**, B. Ng, C. Benoist, S. Mostafavi, Immunological Genome Project. *Proceedings of the National Academy of Sciences of the United States of America*, 2020 [[Link](#)]

*Equal Contribution

INDUSTRY RESEARCH EXPERIENCE **Research Intern**
ServiceNow AI Research *June 2025 - Oct 2025*

- Co-designed DRBench, a benchmark for evaluating AI agents on complex, open-ended deep research questions in enterprise settings.
- Improved DRBench evaluation pipeline and scoring methodology used for model comparison.
- Utilized prompt engineering to finetune inputs to large language models, improving the reliability and accuracy of the model outputs.

ACADEMIC
RESEARCH
EXPERIENCE

Graduate Research Assistant

University of British Columbia - Computer Science

Sept 2021 - Present

- Designed and implemented neural network experiments validating theoretical next-token prediction capacity of transformer models.
- Introduced a theoretical framework unifying local and global smoothness assumptions, proving improved convergence guarantees for different algorithms and step size selection methods.
- Demonstrated experimentally on neural networks that nonmonotone line-search methods operate at the Edge of Stability and enable training with larger step sizes than standard monotone methods.
- Authored master's thesis on the Edge of Stability phenomenon in deep learning.

NSERC Undergraduate Research Assistant

University of British Columbia - Statistics

May 2019 - Aug 2019

- Used convolutional neural networks to identify biologically significant base-pair sequences in genomic data.

NSERC Undergraduate Research Assistant

University of British Columbia - Computer Science

May 2018 - Aug 2018

- Investigated boosting algorithms, summarized in a technical report.

SOFTWARE
EXPERIENCE

Software Developer

Synic Software

2020 - 2021

- Built websites using .NET framework.

TEACHING
EXPERIENCE

Teaching Assistant

University of British Columbia

2015 - 2019, 2021 - 2024

- Courses included machine learning, optimization, algorithms, probability, statistics, and computer systems.

TECHNICAL
SKILLS

Programming Languages: Python, MATLAB, Java, Bash

Packages/Tools: PyTorch, NumPy, Pandas, Matplotlib, Hugging Face, Weights and Biases